CONFIDENTIAL **MOCK EXAM 2026**

GENERIC UNIVERSITY FINAL EXAMINATION

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages

PART A (80 MARKS)

QUESTION 1

a) Using Caesar Cipher technique, decrypt the following ciphertext with a shift of 7 (replace each letter with the one 7 positions to the left).
KLCPJLYLABYULKHUKDPWLKVUL

(2 marks)

b) [THEORY PLACEHOLDER - classical ciphers, 6 marks, Describe. No sourced question passed verification.]

(6 marks)

QUESTION 2

a) List two transposition cipher techniques covered in the lecture and, for each, state how the cipher-text is produced from the plain-text.

(2 marks)

b) Decrypt the following ciphertext using the Vigenere cipher with the key CIPHER. Use the standard 26 x 26 Vigenere tableau, writing the key over the message and repeating it as necessary.
UMRBVVNWVPREQE

(6 marks)

Vigenere tableau (A = 0)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

QUESTION 3

a) State TWO (2) facts about the chain of trust in the Trusted Computing architecture, including how measurements are recorded and where the chain begins.

(2 marks)

b) State SIX (6) facts about the two primary system components of the Next Generation Secure Computing Base (NGSCB), covering the Nexus and the Nexus Computing Agent (NCA).

(6 marks)

QUESTION 4

a) Explain two ways in which the Chinese Wall model differs from traditional access control policies, referring to the type of concern it addresses and how access rights behave over time.

(2 marks)

b) [COMPUTED PLACEHOLDER - acm, 6 marks]

(6 marks)

QUESTION 5

a) List the TWO (2) stages that make up the authentication process in a biometric system.

(2 marks)

b) A bank plans to deploy biometric authentication at its branch counters. Explain the different categories of biometric technologies and the practical issues that must be considered when choosing a biometric mechanism for this deployment.

(6 marks)

QUESTION 6

a) State TWO (2) typical locations where network-based IDS sensors are placed in a network.

(2 marks)

b) Explain six (6) preventive network security mechanisms that an organization can deploy to protect its network, describing for each mechanism how it contributes to preventing or limiting security attacks.

(6 marks)

QUESTION 7

a) State TWO (2) characteristics of a Trojan horse as a type of malware.

(2 marks)

b) Describe six (6) techniques used by malware to propagate itself or to conceal its presence on an infected system, and explain the impact of each technique on the victim.

(6 marks)

QUESTION 8

a) Explain how a buffer overflow arises in a program such as the web server function that copies input into a fixed-size local buffer, and state what an attacker can achieve by exploiting it.

(2 marks)

b) Windows is often described as offering rich built-in security, yet it remains a frequent target of attack. Explain six aspects of Windows security mechanisms, covering the overall nature of these mechanisms, the directory service for organising resources, how subjects and their credentials are handled, the registry's security role, the historical security milestones, and the risk that complexity itself creates.

(6 marks)

QUESTION 9

a) [THEORY PLACEHOLDER - attacks/malware, 2 marks, Explain. No sourced question passed verification.]

(2 marks)

b) Discuss the Samy worm as a case study of a self-propagating web malware attack. In your answer, explain how the worm injected and spread its payload, how victims were drawn into triggering the malicious script, and what the outcome of the outbreak was.

(6 marks)

QUESTION 10

a) Explain two password security controls that an administrator can apply to password files and accounts in order to strengthen password-based authentication.

(2 marks)

b) Refer to your answer in (a). Based on the scenario in (a), explain how Kerberos, smartcards and single sign-on mechanisms can be used to authenticate users across a large distributed environment, giving six distinct points.

(6 marks)

PART B (20 MARKS)

QUESTION 1

A junior developer at a healthcare software firm discovers that a colleague has been copying patients' records into a personal spreadsheet to speed up testing. When challenged, the colleague argues that no written rule forbids it and that the data is only used internally. The firm's management asks the security team to clarify how morals, ethics and laws apply before deciding on any action.

a) Explain how morals, ethics and laws differ as guides for judging the colleague's conduct, and then explain how the ACM code would direct the developer's professional behaviour in this situation.

(5 marks)

b) State five points about the professional code of ethics and conduct that the security team should present to management in the scenario, covering what such a code is, who defines it, an example of a professional body that issues one, and two limitations of relying on it.

(5 marks)

QUESTION 2

A hospital's file server stores patient records and administrative documents. The security team is comparing how access rights could be recorded for each file and each user, and wants a structure that is easy to verify and to manage as staff change roles.

a) Explain the access control structures available for implementing access policies, covering the access control matrix, access control lists and capabilities, and state the requirements such structures must satisfy.

(5 marks)

b) Discuss the goals that authorization aims to achieve and the role of the reference monitor in making authorization decisions, relating your discussion to the hospital file server in the scenario.

(5 marks)

END OF QUESTION PAPER